Guowei HUAI

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EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) — Sept. 2024 – Present
MPhil in Robotics and Autonomous Systems (ROAS) — *GPA: 4.04/4.30* — Guangzhou, China

Beijing Institute of Technology, Zhuhai — Sept. 2019 – June 2023
B.Eng. in Computer Science — *GPA: 89.8/100* — Zhuhai, China

ROBOTICS PROJECTS

Goal-Conditioned In-Hand Rotation Teleoperation Framework — 2025.12 – Present
HKUST(GZ) & ETH Zurich | Project Leader

- Built a two-stage RL training pipeline in IsaacLab: first learning force-closure grasp poses, then using them as initial states to train a multi-axis in-hand rotation policy with tactile feedback for stable object rotation on Sharpa Hand
- Designed a goal-conditioned teleoperation interface that maps tracked in-hand object pose and human-robot hand-joint consistency constraints into policy-level rotation goals, bridging high-dimensional human input with the learned low-level controller
- Built real-world object-pose tracking and rollout/data-collection components for tool-use teleoperation, enabling safe collection of contact-rich in-hand rotation demonstrations under hand-object occlusion and morphology mismatch

HiFun: Hierarchical Real-World RL for Functional Dexterous Manipulation — 2025.05 – 2026.05
HKUST(GZ), HKUST & ETH Zurich | Core Contributor

- Contributed to a hierarchical real-world RL pipeline that decomposes high-DoF arm-hand control into contact-aware hand skills and skill-level coordination, addressing inefficient exploration and timing-sensitive contact failures in functional dexterous manipulation
- Achieved 90–100% success on four real-world dexterous tasks within 20–60 min online RL training, covering power-drill actuation, pipette dispensing, hose-nut unthreading, and constrained tool retrieval
- Evaluated BC, diffusion-policy, HIL-RL, retargeting, and replay baselines on Franka Research 3 with XHand and Sharpa hands, analyzing timing errors, contact instability, demonstration noise, and failure cases under perturbations

Dexterous Teleoperation and Multimodal Data Platform — 2025.05 – 2026.03
HKUST(GZ) | Project Leader

- Captured human hand motion from Vision Pro keypoints and Manus glove signals, implemented optimization-based retargeting with vector- and fingertip-level objectives, and adapted it to Inspire Hand, Leap Hand, XHand, and Sharpa for cross-embodiment teleoperation and demonstration collection
- Implemented Franka arm teleoperation through isomorphic leader-follower GELLO control, 3D mouse control, and Vision-Pro-based end-effector control
- Integrated piezoresistive, electromagnetic, and vision-based tactile sensors into the arm-hand teleoperation stack, synchronizing tactile streams with robot states and demonstrations to build multimodal datasets for real-world RL and imitation learning

Diffusion-Based Visual Imitation Learning for Robotic Manipulation — 2025.03 – 2025.05
HKUST(GZ) Course Project | Contributor

- Reproduced DP3 and RDT-1B on an Airbot Play 6-DoF arm with a parallel gripper, isomorphic teaching arm, external RealSense D455 RGB-D camera, and wrist-mounted RealSense D435 RGB camera
- Processed 490 Airbot500 teleoperation episodes with 3D workspace cropping, farthest-point sampling, and zarr conversion; analyzed real-robot results including 72% seen-task success and 30% unseen-container generalization

INDUSTRY EXPERIENCE

Amicro Semiconductor Co., Ltd. 2023.07 – 2024.06

Algorithm Engineer Zhuhai, China

- Developed perception modules within a home-cleaning robot stack, fusing IMU, LiDAR, line-laser, and RGB camera signals for small-obstacle detection and real-time avoidance
- Optimized YOLOv5-seg for cable-like and low-profile obstacle segmentation on a 1.5 TOPS edge chip, using quantization and hardware-aware deployment to improve inference from 4 fps to 8 fps

PUBLICATIONS & ACADEMIC SERVICE

HiFun: Hierarchical Real-World RL for Functional Dexterous Manipulation, *CoRL 2026 under review, second author* 2026.06

Reviewer, ICRA 2026 Workshop on Manipulation Robustness 2026.05

Method for Obtaining Edge Contours for Robot Navigation, *invention patent* 2024.05

IoT-Based Escape and Evacuation Indication Device, *utility model patent* 2022.09

Motion Image Deblurring Demonstration System, *software copyright* 2022.02

Multifocus Image Fusion Demonstration System, *software copyright* 2022.02

HONORS AND AWARDS

First Prize, HKUST(GZ) Bionic Robot Innovation Competition, MoSense Team (50,000 RMB) 2025.10
Quarterfinalist (Top 8), 1st WBCD Competition @ ICRA 2025 2025.05
HKUST(GZ) MPhil Full Scholarship, RBM Program (240,000 RMB over 2 years) 2024.09
Principal's First-Class Scholarship, BITZH (30,000 RMB; 1/2000) 2023.04
Third Prize, JD Programming and Innovative Application Competition 2022.06
First Prize, Chinese Teddy Cup Data Analysis Competition 2021.12
Climbing Plan Science and Technology Innovation Project Award (30,000 RMB) 2021.05
Outstanding Scholarship, Computer Academy, BITZH 2021.03

SKILLS

Programming & AI Agents : AI Agents(Claude Code, Codex), Git Workflow, PyTorch, JAX
Robot Learning & Robotics : dexterous manipulation, RL/IL, diffusion policy, ROS, Isaac Gym/Lab
Languages : English (IELTS Reading 7.5, Writing 6.5; CET-4/6)